5. **High corrosion resistance:** Aluminium has excellent corrosion resistance and it can resist weathering actions in extremely humid and dry conditions.
6. **High scrap value:** The scrap value of aluminium is very high.
7. **Maintenance cost:** Due to high corrosion resistance, its maintenance cost is very low.
8. **Noise control:** Aluminium is an excellent reflector of electromagnetic and sound waves. It is less affected by external noises as compared to buildings made from other materials.

2.7 Steel

Steel is an alloy of iron, carbon and other elements. Because of its high tensile strength it is a major component used in buildings, infrastructures, tools, ships, automobiles, machines, appliances and weapons. Iron is the base metal of steel.

The important market forms of steel used in building construction are as follows.

1. Steel bars of many shapes and grades or strengths. (These bars are used for reinforced concrete and also for fabrication of grills, gates, etc.)
2. High tensile steel for pre-stressed concrete work.
3. Various shapes of I, channel, angle, plates and other rolled section for structural fabrication.
4. Cold formed light gauge structural steel sections.
5. Stainless steel for special uses.

2.7.1 Types of steel reinforcement

Steel rods used for reinforced concrete work should be of specified tensile strength and they should develop good bond with concrete. There are different types of steel like mild steel, Tor steel, TMT bars, etc. Steel rods of different diameter are used for R.C work.





The following types of bars are commonly available in market for reinforced concrete construction.

1. Hot rolled steel bars
   - i. Hot rolled plain round mild steel bars.
   - ii. Hot rolled ribbed mild steel bars (generally not recommended for use).
   - iii. Hot rolled high strength deformed (HYSD) bars.

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